

Organs as Attractors: A Non-Darwinian Account of Biological Complexity

Claudio Bresciano

Independent Researcher

Theoretical Biology & Axiomatic Necessity Theory (TNA)

Abstract

The standard Darwinian framework explains biological change as the cumulative result of random variation filtered by selection. While effective for incremental optimization within stable regimes, this framework fails to account for the abrupt emergence of complex organs and body plans observed in events such as the Cambrian Explosion and repeated cases of convergent organogenesis (e.g., eyes).

This paper proposes an alternative, necessity-based model in which organs emerge as **morphogenetic attractors** imposed by boundary conditions. Under this view, complex organs are not gradually constructed through stochastic search, but arise via **discontinuous convergence** when minimal coherence thresholds are crossed. Darwinian selection is thus reinterpreted as a local optimizer operating only within structurally stable regimes.

1. Introduction

One of the central unresolved problems in evolutionary theory is the origin of complex organs. Structures such as eyes, lungs, nervous systems, or articulated limbs appear in the fossil record abruptly, fully functional, and repeatedly across unrelated lineages.

Standard explanations invoke:

- gradual accumulation of micro-adaptations,
- incomplete fossil records,
- or extended evolutionary synthesis mechanisms.

However, these explanations rely on an **implicit continuity assumption** that has never been formally demonstrated.

This paper challenges that assumption.

2. The Limits of Darwinian Constructionism

Darwinian evolution models biological novelty as the result of:

1. Random mutation
2. Incremental functional advantage
3. Long-term accumulation

This framework works well for:

- allele frequency shifts,
- fine-tuning of existing traits,
- optimization under stable boundary conditions.

It fails when:

- the configuration space is combinatorially explosive,
- intermediate states are non-functional,
- boundary conditions shift abruptly.

Complex organs fall squarely into this failure domain.

3. Boundary Conditions and Regime Transitions

Biological systems operate under **boundary conditions**: environmental, energetic, geometric, and informational constraints.

When these conditions change gradually, Darwinian dynamics apply.

When they change discontinuously (e.g., oxygenation, radiation, predation, mobility), the system enters a **non-derivable regime** in which:

- Most configurations become non-viable.
- Incremental search collapses.
- Only a small subset of structurally coherent solutions remain.

This is not optimization.
It is **constraint-induced collapse**.

4. Organs as Morphogenetic Attractors

4.1 Definition

A **morphogenetic attractor** is a stable organizational state toward which biological systems converge when constrained by specific boundary conditions.

Properties:

- Low-dimensional solution space
- High functional coherence
- Independence from historical contingency

Organs such as eyes are not arbitrary outcomes—they are **inevitable solutions** under certain physical and informational constraints.

4.2 The Eye as a Canonical Attractor

The eye emerges repeatedly across evolutionary history because:

- Photochemistry is unavoidable in illuminated environments.
- Optical focusing obeys universal geometry.
- Sensory anticipation becomes a minimal coherence requirement under mobility and predation.

The result is ultra-rapid convergence to functional vision systems once boundary thresholds are crossed.

No extended stochastic search is required.

5. Ultra-Rapid Convergence and the Fossil Record

The apparent “explosiveness” of events like the Cambrian Explosion is not a paleontological artifact.

It reflects:

- regime transitions,
- collapse of viable morphospace,
- rapid convergence to attractor states.

From a geological perspective, these transitions appear instantaneous because the limiting factor is not evolutionary time, but **developmental reconfiguration**.

6. Relation to Empirical Boundary-Shift Events

Analogous behavior is observed in:

- Radiation-exposed fauna (e.g., Chernobyl ecosystems),
- Extremophiles,
- Rapid phenotypic expansion in modern Homo sapiens under improved boundary conditions.

In all cases:

- systems do not gradually adapt,
- incoherent configurations collapse,
- coherent regimes persist.

This is selection by **necessity**, not by optimization.

7. Formal Implication: The End of Global Darwinism

Darwinian evolution remains valid as a **local approximation** within stable regimes.

It cannot:

- generate new organs,

- compute new boundary conditions,
- explain discontinuous morphogenesis.

Complexity arises not from chance, but from **structural inevitability**.

8. Conclusion

Biological complexity does not emerge because nature searches blindly until it gets lucky. It emerges because, under constraint, **only certain forms are possible**.

Organs are not inventions of evolution.
They are attractors imposed by reality.

Darwin explains erosion.
This paper explains earthquakes.

References (Representative)

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Final sentence (usable también como cierre fuerte):

Trying to explain the Cambrian Explosion through incremental mutation is like arranging candles and expecting lightning: organs do not assemble—they crystallize when boundary conditions demand them.